

Deep Color-Normal Residual Networks for Geometry Refinement Extracting Color Consistency and Fine Geometry

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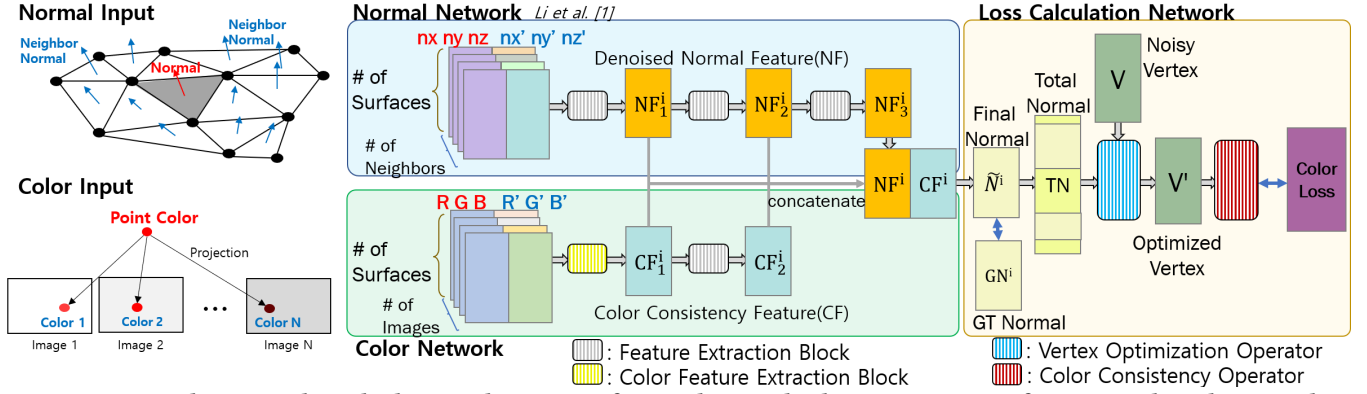


Figure 1: Deep color-normal residual networks consist of normal net and color net: Inputs are facet normals and point colors with color intensities obtained by projecting the points onto color images.

ABSTRACT

In recent years, texture mapping in 3D modeling has been remarkably improved for realistic rendering. However, small error in reconstructed 3D geometry makes serious error in texture mapping. To address the problem, most of prior methods are devoted to refinement of 3D geometry without visual clues. In this work, we refine 3D geometry based on color consistency and surface normal using a deep neural network. Our method optimizes the location of each vertex maximizing the quality of related textures.

CCS CONCEPTS

• **Computing methodologies** → *Reconstruction*.

KEYWORDS

geometry refinement, reconstruction, deep neural network

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1 INTRODUCTION

Modeling 3D objects from multi-view images with/without depth data is an interesting but challenging task. Systematic and non-systematic errors in the process of 3D reconstruction such as feature matching cause limited performance. One approach to address the problem is geometry refinement. Geometry refinement is a method of improving 3D shapes including the aforementioned errors. [Xie et al. 2017] use angle profile defined from normal maps to enhance 3D surface details that are lost because of sampling and quantization. To remove the noise of the 3D shape, [Li et al. 2020] propose a denoising network that acts as a normal filter. They feed the surface normal and its neighbors to the network and get smoothed normal. While most studies of geometry refinement help to reduce geometric errors, they improve reconstructed geometry ignoring the quality of texture. Improving the geometry without considering texture makes the reconstructed object unrealistic and causes seam artifacts. In this paper, we optimize 3D shape details based on not only geometry but also texture. We propose deep color-normal residual networks simultaneously extracting both surface normal and color consistency. Inspired by [Li et al. 2020], we compute the residual of point color and color intensities obtained by projecting a 3D point to related images. Color residual helps the network to optimize global and local color consistency [Wang et al. 2019].

2 PROPOSED METHOD

2.1 Network Architecture

Our network structure is based on [Li et al. 2020] which consists of feature extraction and denoising parts. In our training, we adopt the feature extraction part to learn surface normal feature representations expecting denoised surface normal output as well (Normal

Network in Figure 1). Unlike the inputs of normal network, inputs of color network (in Figure 1) are point colors and color intensities obtained by projecting the point onto related images. Feature extraction block in the color network extracts color consistency of a reference surface and its neighboring surfaces in a 3D Euclidean space. While general feature extraction blocks of normal(or color) network extract residual normal(or color) features from neighbor region, color feature extraction block of color network extracts residual color features from point projected images. In other words, color feature extraction block outputs residual features between point color and color intensities with its projection (Color Input in Figure 1). Finally, we concatenate all normal and color features and regress to generate final predicted normal $\tilde{N}^i$.

2.2 Loss Function

We calculate L2 norm between predicted $\tilde{N}^i$ and ground-truth normal GN^i (geometry loss) and minimize the difference to let our network to extract smoother surface normal. Simultaneously with the geometry loss, we iteratively update noisy vertices V with predicted normal $\tilde{N}^i$ using [Sun et al. 2007] in the vertex optimization operator:

$$\mathbf{x}' = \mathbf{x} + \frac{1}{|F_V|} \sum_{k \in F_V} \mathbf{n}'_k (\mathbf{n}'_k \cdot (\mathbf{c}_k - \mathbf{x})), \quad (1)$$

where $\mathbf{x}'$ is updated vertex position of V , F_V is the set of faces that share the common vertex V , $\mathbf{c}_k$ is the center of the triangle and $\mathbf{n}'_k$ is the denoised normal $\tilde{N}^i$ of the surface. By projecting the optimized vertices V' to images, we calculate color consistency between the images (color loss). Since calculating the color consistency is not differentiable function, it cannot indicate learning direction to correct vertex position in the back-propagation step. Alternatively, we propose to sum Gaussian jitter value to current point position of V' and calculate color loss with the jittered points. Finally, we perform back-propagation following the jitter value that calculates the lowest loss value. We encode this into loss function as follows.

$$\begin{aligned} Loss_{col} &= \frac{1}{N} \sum_{k \in images} (C(\mathbf{x}' + \mathbf{t}_i) - I_k(\pi(\mathbf{x}' + \mathbf{t}_i))) \\ \nabla \mathbf{x}' &= \underset{\mathbf{t}_i}{\operatorname{argmin}} (Loss_{col}(\mathbf{x}' + \mathbf{t}_i)), \quad \mathbf{t}_i \in \mathbb{R}^3 \end{aligned} \quad (2)$$

where $C(x)$ means point color of x , $\pi(x)$ is a function that projects point x onto image and returns a pixel coordinate on the image, $I_k(u)$ is a RGB value at u in image k and $\mathbf{t}_i$ is a jitter value. The final $Loss_{total}$ is defined as follows.

$$Loss_{total} = \alpha Loss_{geo} + Loss_{col}, \quad (3)$$

where α is a weight to balance the relative influence of the two loss terms. $Loss_{geo}$ and $Loss_{col}$ indicate geometry loss and color loss respectively. The network learns to extract color consistency features and the predicted normal $\tilde{N}^i$ moving the noisy vertices V toward the right direction. Even though color loss is reduced and the points move toward the right direction, each point may independently back-propagate gradients resulting in messy geometry. However, geometry loss suppresses the movement of points maintaining the shape of the smoothed surface.

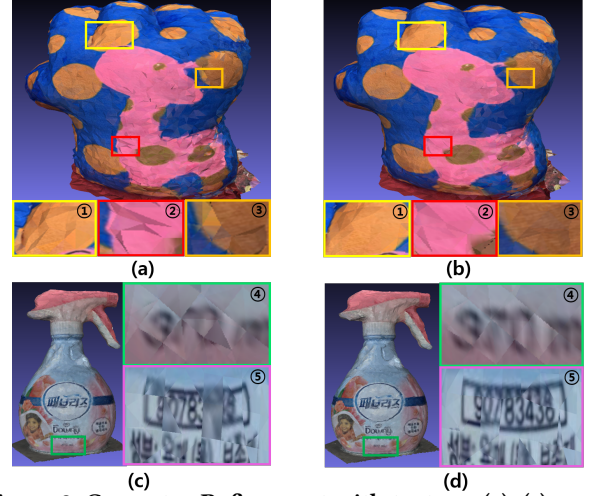


Figure 2: Geometry Refinement with texture: (a), (c) are before refinement and (b), (d) are after refinement.

3 EXPERIMENTAL RESULTS

Evaluation is performed on our dataset obtained by a structure-from-motion (SfM). The dataset includes noisy 3D mesh models. Normal network uses weights of pre-trained network of [Li et al. 2020]. As shown in Figure 2 (a), (b), our network makes 3D mesh smoother and removes noise by minimizing geometry loss. In Figure 2 (a)-1 and (a)-2, object surface looks rough due to geometric noise resulting in shadows observed on the surface. However, in (b)-1 and (b)-2, the noise is removed and the surface looks smoother and cleaner. In (a)-3, texture color errors are observed due to geometry errors. But in (b)-3, it looks smoothed and uniform color is mapped. If the influence of the geometry loss term becomes weaker by reducing α in equation (1), geometric noise becomes serious. As shown in Figure 2(c), (d), our method improves geometry details by employing texture clue that could not be improved with only geometry clue. Text on the surface looks distorted due to geometry errors in (c)-4. However, (d)-4 shows more clear and legible text. The line of the square box is distorted in (c)-5, but it is well aligned in (d)-5.

4 CONCLUSION

We propose deep geometry refinement with local color consistency and geometry jointly. Our method refines 3D geometry complementarily with geometry loss and texture loss.

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